

Solutions to JEE Advanced – 2 | JEE 2023 | Paper-1 (Gen 1 & 2)

PHYSICS

1.(C) Since A moves faster than B, clearly it will cover a greater distance before they meet

So, we can look at the situation as A being three-quarters of the circle, i.e. a distance $\frac{3\pi R}{2}$ behind B initially.

So, the particles will be at diametrically opposite points on the circle when A is a half-circle behind B. Hence, the time instant when this happens is given by

$$s_A = s_B + \frac{\pi R}{2}$$

$$vt_1 = \left(\frac{2v}{3}\right)t_1 + \frac{\pi R}{2}$$

$$\Rightarrow t_1 = \frac{3\pi R}{2v}$$

Now, the distance travelled by the particles until $t = \frac{3\pi R}{2v}$ is

$$s_A = v\left(\frac{3\pi R}{2v}\right) = \frac{3\pi R}{2} \quad \text{and} \quad s_B = \left(\frac{2v}{3}\right)\left(\frac{3\pi R}{2v}\right) = \pi R$$

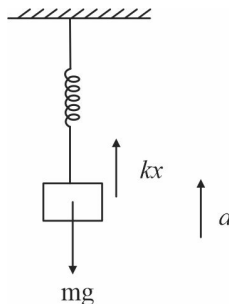
So, at $t = \frac{3\pi R}{2v}$, the coordinates of A and B are respectively $(0, R)$ and $(0, -R)$

Hence, their average velocity in the interval t_1 is respectively

$$\frac{\sqrt{2}R}{\left(\frac{3\pi R}{2v}\right)} = \left(\frac{2\sqrt{2}}{3\pi}\right)v \quad \text{and} \quad \frac{2R}{\left(\frac{3\pi R}{2v}\right)} = \left(\frac{4}{3\pi}\right)v$$

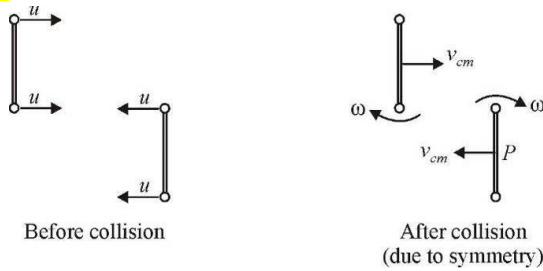
2.(D) After the block has moved downwards by a distance x , the net force it on upward direction,
 $F = kx - mg$

Therefore, its acceleration, $a = \frac{F}{m} = \left(\frac{k}{m}\right)x - g$



We can see that the graph must be a straight line with positive slope, and the value of a at $x = 0$ is $-g$

3.(C)



$$V_{sep} = e V_{app}$$

$$\left(\frac{\omega l}{2} - V_{cm} \right) \times 2 = 2u$$

$$\Rightarrow \omega l - 2V_{cm} = 2u \quad \dots (i)$$

Conserving angular momentum about the centre of mass of the rod B (taking clockwise positive)

$$\left(mu \frac{l}{2} + mu \times \frac{3l}{2} \right) + \left(mu \times \frac{l}{2} - mu \frac{l}{2} \right) = \left(\frac{ml^2}{4} \right) \times 2 \times \omega + \left[\left(2 \times \frac{ml^2}{4} \times \omega \right) + 2m \times V_{cm} \times l \right]$$

$$\Rightarrow 2mul = ml^2\omega + 2mV_{cm}l \quad \Rightarrow \quad 2u = \omega l + 2V_{cm} \quad \dots (ii)$$

From (i) and (ii), we get $\omega = \frac{2u}{l}$

4.(B) The horizontal component of the velocity of the ball remains unchanged throughout.

$$\text{So, } u \sqrt{\frac{2h_1}{g}} + u \sqrt{\frac{2h_2}{g}} = L \quad \Rightarrow \quad h_2 = \left(\frac{L}{u} \sqrt{\frac{g}{2}} - \sqrt{h_1} \right)^2$$

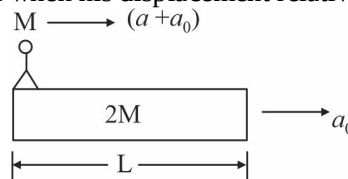
5.(A) Let the acceleration of the plank relative to the ground be a_0

Then, the acceleration of the man relative to the ground is $a + a_0$

Since the external force on the system is zero, the acceleration of the CM must be zero

$$\text{Therefore, } a_{CM} = \frac{M(a + a_0) + 2M(a_0)}{M + 2M} = 0 \quad \Rightarrow \quad a_0 = -\frac{a}{3}$$

Now, the man reaches the other end when his displacement relative to the plank has become L



$$\text{Therefore, } s_{rel} = \frac{1}{2}(a)t^2 = L \quad \Rightarrow \quad t = \sqrt{\frac{2L}{a}}$$

$$\text{So, the velocity of the plank at this instant is } v = -\left(\frac{a}{3}\right)t = -\frac{\sqrt{2}}{3}\sqrt{aL}$$

6.(B) If F is increased beyond a limit, block D will slip over C. This is because friction force acting on D is responsible to acceleration D + B + A. Hence, for all the four block to move together –

Acceleration of the system produced by $F \leq$ acceleration produced by friction on D in the system D + B + A

$$\therefore \frac{F}{2m_1 + 2m_2} \leq \frac{\mu m_2 g}{m_1 + 2m_2} \quad \therefore \quad F_0 = 2\mu m_2 g \left(\frac{m_1 + m_2}{2m_2 + m_1} \right)$$

7.(ABCD)

Equation of trajectory of the projectile:

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

Since the particle must pass through the point (15, 5),

$$5 = 15 \tan \theta - \frac{10(15)^2}{2(15)^2} (1 + \tan^2 \theta)$$

Simplifying, we get a quadratic equation in $\tan \theta$:

$$\tan^2 \theta - 3 \tan \theta + 2 = 0$$

So, the two solutions are: $\theta = 45^\circ$ and $\theta = \tan^{-1} 2$

Now, the time taken for the projectile to hit the point P can be simply found by using the motion in the X-

direction: $(u \cos \theta)t = 15 \Rightarrow t = \frac{15}{15 \cos \theta} = \frac{1}{\cos \theta}$

So, the time in the two cases is: $\frac{1}{\cos 45^\circ} = \sqrt{2} \text{ s}$ and $\frac{1}{\cos(\tan^{-1} 2)} = \sqrt{5} \text{ s}$

Since we are given that $t_1 < t_2$, we get

$$t_1 = \sqrt{2} \text{ s} \quad \text{and} \quad t_2 = \sqrt{5} \text{ s}$$

$$\theta_1 = 45^\circ \quad \text{and} \quad \theta_2 = \tan^{-1} 2$$

8.(AD) Since the velocity of the bottom-most point at $t = 0$ is towards right, kinetic friction $f_k = \mu mg$ acts towards left

Therefore, the acceleration of the centre of mass,

$$a_{CM} = \frac{f_k}{m} = \mu g \quad (\text{towards left})$$

And, the angular acceleration of the cylinder,

$$\alpha = \frac{f_k R}{\frac{1}{2} m R^2} = \frac{2 f_k}{m R} = \frac{2 \mu g}{R} \quad (\text{clockwise})$$

So, taking rightward positive and clockwise positive, at any general time instant t ,

$$v_{CM} = v_0 - \mu g t \quad \text{and} \quad \omega = -\omega_0 + \left(\frac{2 \mu g}{R} \right) t$$

Pure rolling begins when $v_{CM} = R\omega$

(Note carefully that we can write this because we have taken rightward positive for v_{CM} and clockwise positive for ω)

$$\text{So, } v_0 - \mu g t = R \left(-\omega_0 + \left(\frac{2 \mu g}{R} \right) t \right) \Rightarrow t = \frac{v_0 + R\omega_0}{3 \mu g} = \frac{1 + (0.1)(50)}{3(0.1)(10)} = 2 \text{ s}$$

Therefore, at $t = 2 \text{ s}$,

$$v_{CM} = v_0 - \mu g t = 1 - (0.1)(10)(2) = -1 \text{ m/s}$$

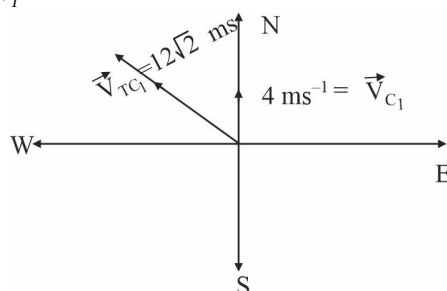
$$\text{So, } v_{CM} = 1 \text{ m/s towards left and, } \omega = -\omega_0 + \left(\frac{2 \mu g}{R} \right) t = -50 + \left(\frac{2(0.1)(10)}{0.1} \right) (2) = -10 \text{ rad/s}$$

So, $\omega = 10 \text{ rad/s}$ anti-clockwise

Once pure rolling begins, v_{CM} and ω remain constant at the values they had at $t = 2 \text{ s}$.

9.(ABCD)

Let the positive X-axis be towards East and the positive Y-axis towards North
Let the velocity of the train be $\vec{v}_T$



Let the initial velocity of the cyclist be $\vec{v}_{C1} = 4\hat{j}$

Let the initial velocity of the train relative to the cyclist be $\vec{v}_{TC1}$

We are given that $\vec{v}_{TC1} = -12\hat{i} + 12\hat{j}$

Now, we know from the definition of relative velocity that $\vec{v}_{TC1} = \vec{v}_T - \vec{v}_{C1}$

Therefore, $\vec{v}_T - \vec{v}_{C1} = -12\hat{i} + 12\hat{j}$

So, $\vec{v}_T = \vec{v}_{C1} + (-12\hat{i} + 12\hat{j}) = -12\hat{i} + 16\hat{j}$

Hence, the train's actual velocity is 20 m/s, in a direction $\tan^{-1}\left(\frac{12}{16}\right) = \tan^{-1}\left(\frac{3}{4}\right)$ West of North

Now, if the cyclist moves towards East, his velocity is $\vec{v}_{C2} = 4\hat{i}$

Therefore, the velocity of the train as seen by him now is $\vec{v}_{TC2} = \vec{v}_T - \vec{v}_{C2} = (-12\hat{i} + 16\hat{j}) - 4\hat{i} = -16\hat{i} + 16\hat{j}$

The velocity of the train as seen by the cyclist is $\vec{v}_{TC} = \vec{v}_T - \vec{v}_C$

So, it is clear that if he orients his velocity opposite to the train's actual velocity, their vector subtraction has the maximum magnitude

10.(BCD)

String becomes tight again when distance travelled by the two blocks becomes equal.

$$\Rightarrow 6t = 2t + \frac{1}{2}gt^2 \Rightarrow t = 0.8s$$

$$\text{At } t = 0.8s, \quad v_1 = 6m/s$$

$$v_2 = 2 + (0.8)g = 10 m/s.$$

As string becomes tight string applies equal impulse on the blocks in the directions shown.

$\Rightarrow$ We can equate change in momentum of the blocks.

$$(1)(6) + (1)(10) = (1+1)V$$

$$\Rightarrow v = 8m/s \quad \Rightarrow \quad \text{Blocks move at } v = 8m/s \text{ immediately after } t = 0.8s.$$

So, impulse on block 2 during tightening is $I_2 = 1(-8 - (-10)) = 2 \text{ kg m/s}$

$$\text{Heat generated} = KE_i - KE_f = \left(\frac{1}{2}(1)(6)^2 + \frac{1}{2}(1)(10)^2 \right) - 2 \left(\frac{1}{2}(1)(8)^2 \right) = 4 \text{ J}$$

11.(BD)

$$\begin{aligned} \text{Initial momentum} &= \text{Final momentum} \\ \text{Therefore,} \quad mu &= \sqrt{(mv)^2 + \left(m\frac{v}{2}\right)^2} + 2(mv)\left(m\frac{v}{2}\right)\cos 60^\circ \\ \Rightarrow \quad u &= \frac{\sqrt{7}v}{2} \quad \Rightarrow \quad v = \frac{2u}{\sqrt{7}} \end{aligned}$$

Now, since the common normal is along the final direction of motion of disk B, the angle between the final direction of motion of A and the common tangent is 30°

And, the angle between the initial direction of motion of A and the common tangent is $\theta + 30^\circ$

So, equating the component of A's velocity along the common tangent before and after the collision,

$$u \cos(\theta + 30^\circ) = v \cos(30^\circ)$$

$$\Rightarrow \cos(\theta + 30^\circ) = \frac{\sqrt{3}v}{2u} = \sqrt{\frac{3}{7}}$$

12.(AD)

Torque is the rate of change of angular momentum

Think of torque and angular momentum similar to how you think about force and momentum

During the flight of the particle, the only force acting on it is its weight vertically downward

Clearly, the angular momentum about P is initially zero, and since the torque on the particle (due to the particle's weight) about P is always clockwise, the angular momentum about P keeps increasing

About Q , the angular momentum is anti-clockwise just after projection, and clockwise just before landing. And the torque on the particle (due to the particle's weight) about Q is always clockwise. Hence, we can deduce that the initially anti-clockwise angular momentum about Q must have reduced to zero and then increased in the clockwise direction

1.(280) Let the time taken from the beginning until the crossing be t

$$\text{Then, } \frac{1}{2} \left(a + \frac{a}{2} \right) t^2 = 300 \quad \Rightarrow \quad at^2 = 400$$

The velocities of A and B at the instant of crossing are at and $\frac{at}{2}$ respectively

Therefore, the time taken by the cars to stop after the crossing is respectively

$$\frac{at}{(5a/2)} = 2t/5 \text{ and } \frac{at/2}{a} = t/2$$

$$\text{So, the total time for which the car A moved is } t + \frac{2t}{5} = \frac{7t}{5}$$

$$\text{Since } \frac{7t}{5} = 14 \Rightarrow t = 10 \text{ s}$$

$$\text{Therefore, } a = \frac{400}{10^2} = 4 \text{ m/s}^2$$

So, the speed of A at the instant of crossing is $(4)(10) = 40 \text{ m/s}$ and its deceleration afterwards

$$\text{is } \frac{5(4)}{2} = 10 \text{ m/s}^2$$

$$\text{So, the total distance moved by A is } s_A = \frac{1}{2}(4)(10)^2 + \frac{(40)^2}{2(10)} = 280 \text{ m}$$

2.(15) The total time for which B moved is $t + \frac{t}{2} = 15 \text{ s}$

3.(2.40) 4.(1.20)

Let the duration of the collision be denoted by τ

We know that $F(\tau)$ must be zero

$$\Rightarrow A\tau - B\tau^2 = 0 \quad \Rightarrow \quad \tau = \frac{A}{B}$$

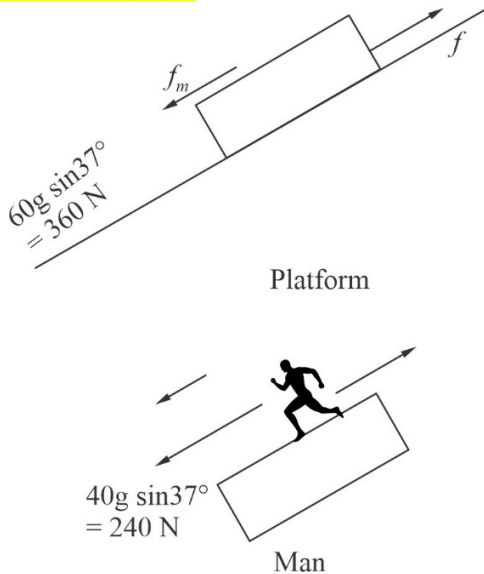
Now, applying impulse momentum theorem along the horizontal direction

$$\int_0^\tau F(t) dt = |\Delta \vec{p}| \quad (F(t) \text{ is the only force acting in the horizontal direction})$$

$$\Rightarrow \frac{A\tau^2}{2} - \frac{B\tau^3}{3} = (0.2)(3 - (-5)) \Rightarrow \frac{A^3}{6B^2} = \frac{8}{5} \Rightarrow \frac{A}{6}\tau^2 = \frac{8}{5}$$

$$\Rightarrow A = \frac{(8)(6)}{5(0.02)^2} = 2.4 \times 10^4 \text{ N/s} \text{ And, } B = \frac{A}{\tau} = \frac{2.4 \times 10^4}{0.02} = 1.2 \times 10^6 \text{ N/s}^2$$

5.(320) 6.(40)



Normal reaction by the incline on the platform is

$$N = 100g \cos 37^\circ = 800 \text{ N}$$

$\therefore$ Maximum possible friction force on the platform (by incline) is

$$f_{\max} = \mu N = 400 \text{ N}$$

The maximum friction by the man on the platform is downward direction can be $f_m = 40 \text{ N}$. This means that the man is experiencing friction force of 40 N up the plane.

$$\text{Acceleration of man in this case } a = \frac{240 - 40}{40} = 5 \text{ m/s}^2$$

If $f_m = 0$; acceleration of man $a = 6 \text{ m/s}^2$

If acceleration of man is large than 6 m/s^2 , the friction f_m reverse direction.

When $a = 7 \text{ m/s}^2$

$$f_m = 40 \text{ N}$$

Friction on platform by incline will adjust itself to 320 N .

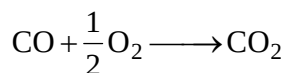
CHEMISTRY

$$1.(C) \quad \begin{array}{cc} \text{CO} & \text{CO}_2 \\ x \text{ mol} & (1-x) \text{ mol} \end{array}$$

$$(\text{eq})_{\text{CO}_2} = (\text{eq})_{\text{NaOH}}$$

$$(1-x) \times 2 = \frac{20}{40} \times 1$$

$$x = \frac{3}{4}$$



$$t = 0 \quad \begin{array}{cc} x \text{ mole} & (1-x) \text{ mol} \\ 0 & x + (1-x) = 1 \text{ mol} \end{array}$$

$$(\text{eq})_{\text{CO}_2} = (\text{eq})_{\text{NaOH}}$$

$$1 \times 2 = \frac{x}{40} \times 1 \Rightarrow x = 80 \text{g}$$

$$\text{Extra amount of NaOH} = 80 - 20 = 60 \text{g} = 1.5 \text{ mole}$$

$$2.(C) \quad V_A = 0.5, \quad V_B = 1$$

$$T_A = T_B$$

$$\frac{d_A}{d_B} = 2, \quad \frac{M_A}{M_B} = \frac{1}{2}$$

$$P = \frac{d}{M} \times RT$$

$$\frac{P_A}{P_B} = \frac{d_A}{d_B} \times \frac{M_B}{M_A} = 2 \times 2 = 4$$

$$3.(D) \quad \frac{P}{V} = \text{const.} \Rightarrow \text{polytropic process}$$

$$PV^{-1} = \text{const.} \Rightarrow x = -1$$

$$w = \frac{nR}{(x-1)} (T_2 - T_1)$$

$$\Delta U = n \frac{3}{2} R (T_2 - T_1)$$

$$\frac{\Delta U}{w} = \frac{3}{2} (x-1) = \frac{3}{2} (-2) = -3$$

$$4.(C) \quad T \propto \frac{n^3}{Z^2}$$

$$\frac{T_3, \text{Li}^{2+}}{T_2, \text{He}^+} = \left(\frac{3}{2}\right)^3 \times \left(\frac{2}{3}\right)^2 = \frac{3}{2}$$

$$T_2, \text{He}^+ = \frac{2}{3} T_3, \text{Li}^{2+} = \frac{2}{3} \times s$$

- 5.(A) (B) It violates Hund's rule
 (C) It violates Aufbau principle
 (D) It violates Aufbau principle

6.(C) $-25 = 4x + y - (3x + 90 + 110)$

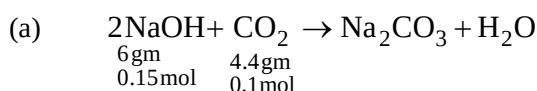
$$x + y = 175$$

$$x = \frac{9}{5} y$$

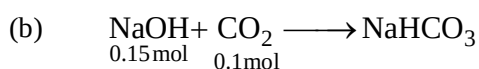
$$\frac{9}{5} y + y = 175 \Rightarrow y = \frac{175 \times 5}{14}$$

$$y = 62.5 \text{ kcal.}$$

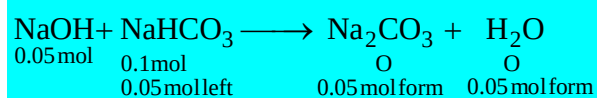
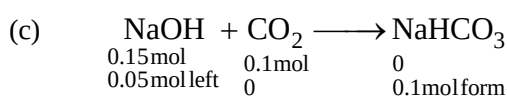
7.(ABC)



NaOH will be limiting reagent



CO₂ will be limiting reagent



Moles of NaHCO₃ left = moles of Na₂CO₃ formed but masses are not equal.

Total mass of reaction mixture

$$= W_{\text{H}_2\text{O}} + W_{\text{Na}_2\text{CO}_3} + W_{\text{NaHCO}_3}$$

$$= 0.05[18 + 106 + 84]$$

$$= 10.4 \text{ gm (Law of mass conservation)}$$

8.(ABD)

Correct order : S > O (electron affinity)

9.(ACD)

- (A) Magnetic moment: $\text{Cr}^{2+} = \text{Co}^{3+}$ (both have 4 unpaired electrons)
 (C) I.P. $\Rightarrow \text{Sn} < \text{Pb}$
 (D) Element belongs to 13th group

10.(BC)

- (A) No. of angular nodes in 3s-orbital = 0
 (D) d_{z²} orbital has non zero electron density in XY-plane

11.(BC)

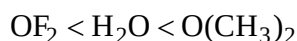
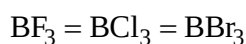
$$\frac{(U_{\text{rms}})_A}{(U_{\text{rms}})_B} = \sqrt{\frac{T_A}{T_B} \times \frac{M_B}{M_A}}$$

$$\frac{0.866}{1} = \sqrt{\frac{300}{400} \times \frac{M_B}{M_A}}$$

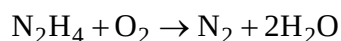
$$\frac{M_B}{M_A} = 0.75 \times \frac{4}{3} = 1$$

12.(BCD)

Correct order of bond angle



1.(-147)



$$\Delta H = \left(-\frac{240}{4}\right) + \left(\frac{36}{4}\right) - \frac{3}{4}(-76) - \frac{9}{4}(68) = -147 \text{ kcal/mol}$$

2.(11) $\text{N}_2(\text{g}) + 2\text{H}_2(\text{g}) \longrightarrow \text{N}_2\text{H}_4(\ell)$

$$\Delta H = \left(\frac{240}{4}\right) - \left(\frac{36}{4}\right) + \frac{3}{4}(-76) + \frac{68}{4} = 11 \text{ kcal/mol}$$

3.(1)

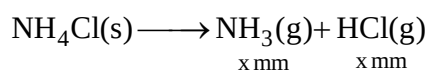
4.(300)

PV = nRT for He gas

$$\frac{190}{760} \times V = 0.01 \times \frac{1}{12} \times 300$$

$$V = 1 \text{ L}$$

$$P \propto T \Rightarrow P_{\text{He}} \text{ at } 600\text{K} = 380 \text{ mm of Hg}$$



$$x + x + 380 = 980$$

$$P_{\text{NH}_3} = x = 300 \text{ mm of Hg} = 300 \text{ Torr}$$

5.(7) PCl_6^- , SF_6 , XeO_3F_2 , SCH_2F_4 , SF_4 , XeO_6^{4-} , I_2Cl_6 6.(5) O_2 , B_2 , O_2^+ , O_2^- , H_2^+ are paramagnetic species

MATHEMATICS

1.(B) Other root will be $2 - \sqrt{3}$. as $a, b, c \in Q$

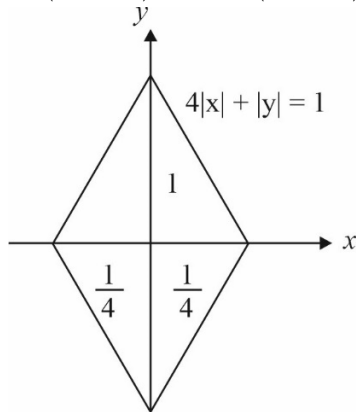
$$\therefore \text{roots} = 2 \pm \sqrt{3}$$

$$\therefore \frac{-b}{a} = 4, \quad \frac{c}{a} = (2 + \sqrt{3})(2 - \sqrt{3}) = 1$$

$$\Rightarrow b = -4a, c = a \Rightarrow \frac{a + 2b + 3c}{a + 3b + 2c} = \frac{-4}{-9} = \frac{4}{9}$$

2.(A) $z\bar{z}^3 + \bar{z}z^3 = 30$

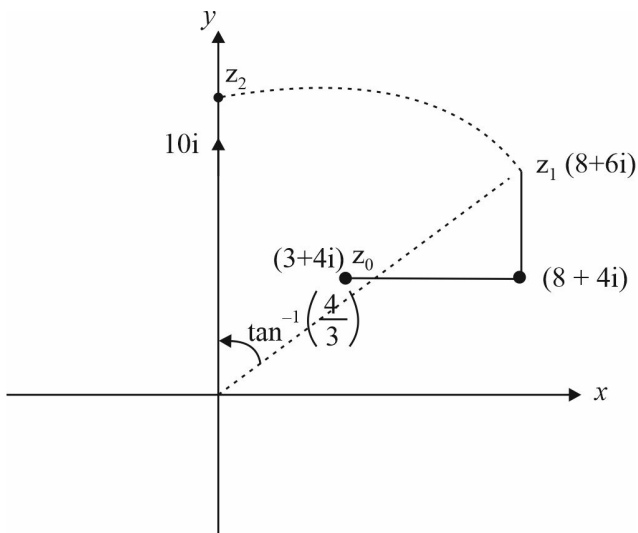
$$\Rightarrow z\bar{z}(z^2 + \bar{z}^2) = 30 \Rightarrow (a^2 + b^2)2(a^2 - b^2) = 30$$



$$\Rightarrow a^2 + b^2 = 5, a^2 - b^2 = 3 \Rightarrow a^2 = 4, b^2 = 1$$

$$\text{Area of rhombus} = 4 \times \left(\frac{1}{2} \times \frac{1}{4} \times 1 \right) = \frac{1}{2}$$

3.(C)



4.(B) $x - 1 \geq 0 \Rightarrow x \geq 1$

$$1 - x \geq 0 \Rightarrow x \leq 1 \Rightarrow x = 1$$

5.(C) Coefficient of x^9 in $(x+1)^5 (x+2)^5$

$$= ({}^5C_0)({}^5C_1 \cdot 2) + ({}^5C_1)({}^5C_0)$$

$$= 10 + 5 = 15$$

6.(B) No. of rectangles including squares

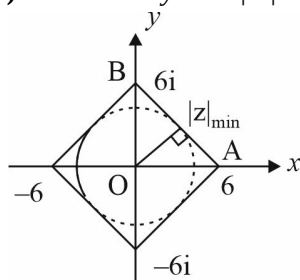
$$= {}^{10}C_2 ({}^{11}C_2)$$

No. of squares only

$$= \sum_{r=1}^9 (11-r)(10-r) = \sum_{r=1}^9 (110 - 21r + r^2) = 110 \times 9 - 21 \left(\frac{9 \times 10}{2} \right) + \frac{9(10)(19)}{6} = 330$$

$\therefore$ No. of rectangles excluding squares = $2475 - 330 = 2145$

7.(ABC) Let $z = x + iy$ then $|x| + |y| = 6$



$$\ar(\Delta OAB) = \frac{1}{2} \times 6 \times 6 = \frac{1}{2} \times 6\sqrt{2} \times |z|_{\min} \Rightarrow |z|_{\min} = 3\sqrt{2} \quad \therefore \sqrt{18} \leq |z| \leq \sqrt{36}$$

8.(ABCD) $x^3 - x + 1 = 0 \rightarrow$ roots : α, β, γ

$$\therefore \alpha + \beta + \gamma = \Sigma\alpha = 0$$

$$\alpha\beta + \beta\alpha + \gamma\alpha = \Sigma\alpha\beta = -1$$

$$\alpha\beta\gamma = -1$$

$$\therefore \alpha^2 + \beta^2 + \gamma^2 = \Sigma\alpha^2 = (\Sigma\alpha)^2 - 2\Sigma\alpha\beta = 2$$

$$\text{Now, } \alpha^3 - \alpha + 1 = 0 \Rightarrow \alpha^3 = \alpha - 1 \Rightarrow \Sigma\alpha^3 = \Sigma\alpha - \Sigma 1 = -3$$

$$\alpha^3 = \alpha - 1 \Rightarrow \alpha^4 = \alpha^2 - \alpha \Rightarrow \Sigma\alpha^4 = \Sigma\alpha^2 - \Sigma\alpha = 2$$

$$\alpha^3 = \alpha - 1 \Rightarrow \alpha^5 = \alpha^3 - \alpha^2 \Rightarrow \Sigma\alpha^5 = \Sigma\alpha^3 - \Sigma\alpha^2 = (-3) - 2 = -5$$

$$\alpha^3 = \alpha - 1 \Rightarrow \alpha^6 = \alpha^2 - 2\alpha + 1 \Rightarrow \Sigma\alpha^6 = \Sigma\alpha^2 - 2\Sigma\alpha + \Sigma 1 = 5$$

$$9.(ABC) \frac{1}{3} \leq \frac{x^2 - x + 1}{x^2 + x + 1} \leq 3 \Rightarrow \log_2 \left[\frac{x^2 - x + 1}{x^2 + x + 1} \right] = \log_2 1, \log_2 2, \log_2 3 = 0, 1, \log_2 3$$

$$\log_2 3 > 1.5 \Rightarrow 3 > 2^{1.5}$$

$$10.(BC) f(-x) = (e^{-x} + e^x) \{ \tan(-x) + \sin^3(-x) \} = -f(x)$$

$$g(-x) = (-x) \log \left(\frac{x^2 + x + 1}{x^2 - x + 1} \right) = x \log \left(\frac{x^2 - x + 1}{x^2 + x + 1} \right) = g(x)$$

$$h(-x) = \sin \{ \cos(-x - \tan x) \} = \sin \{ \cos(x + \tan x) \} = h(x)$$

$$k(-x) = 2^{-x} \neq k(x) \text{ for all } x$$

11.(ABCD) AA DD II M N R V Y

Selection of 5 letters

No. of ways

2 alike 2 alike 1 different

$$({}^3C_2)({}^6C_1) = 18$$

2 alike 3 different

$$({}^3C_1)({}^7C_3) = 105$$

All different

$${}^8C_5 = 56$$

Total

$$179$$

$$\begin{aligned} &\text{Coefficient of } x^5 \text{ in } (1+x+x^2)^3 (1+x)^5 \\ &= \text{coefficient of } x^5 \text{ in } (1+3x+6x^2+7x^3+6x^4+3x^5+x^6) (1+x)^5 \\ &= {}^5C_5 + 3 \cdot {}^5C_4 + 6 \cdot {}^5C_3 + 7 \cdot {}^5C_2 + 6 \cdot {}^5C_1 + 3 \cdot {}^5C_0 = 179 \end{aligned}$$

$$\begin{aligned} &\text{Coefficient of } x^6 \text{ in } (1+x+x^2)^3 (1+x)^5 \\ &= 3 \cdot {}^5C_5 + 6 \cdot {}^5C_4 + 7 \cdot {}^5C_3 + 6 \cdot {}^5C_2 + 3 \cdot {}^5C_1 + {}^5C_0 = 179 \end{aligned}$$

$$\lambda + 3 = 182 = 2 \times 7 \times 13 \rightarrow \text{No. of divisors} = 8$$

$$\lambda + 1 = 180 = 2^2 \times 3^2 \times 5 \rightarrow \text{No. of divisors} = 18$$

$$12.(\text{BCD}) \quad (1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + \dots + {}^nC_n x^n \quad \dots(1)$$

$$(x+1)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} + {}^nC_2 x^{n-2} + \dots + {}^nC_n \quad \dots(2)$$

Multiply (1) and (2) and equate coefficient of x^n both sides, we get

$${}^{2n}C_n = {}^nC_0^2 + {}^nC_1^2 + {}^nC_2^2 + \dots + {}^nC_n^2 = \sum_{r=0}^n {}^nC_r^2$$

$$\text{Put } x = -1 \text{ in (1), to get, } 0 = {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \dots + (-1)^n {}^nC_n$$

$$\text{Put } x = 1 \text{ in (1) to get, } 2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n$$

$$\text{Use } (1+x)^{2n} = {}^{2n}C_0 + {}^{2n}C_1 x + {}^{2n}C_2 x^2 + \dots + {}^{2n}C_{2n} x^{2n}$$

$$\text{Put } x = 1, \quad 2^{2n} = {}^{2n}C_0 + {}^{2n}C_1 + {}^{2n}C_2 + \dots + {}^{2n}C_{2n}$$

$$4^n = ({}^{2n}C_0 + {}^{2n}C_1 + {}^{2n}C_2 + \dots + {}^{2n}C_{n-1})$$

$${}^{2n}C_n + ({}^{2n}C_{n+1} + {}^{2n}C_{n+2} + \dots + {}^{2n}C_{2n-1} + {}^{2n}C_{2n})$$

$$\Rightarrow {}^{2n}C_0 + {}^{2n}C_1 + {}^{2n}C_2 + \dots + {}^{2n}C_{n-1} = \frac{4^n - {}^{2n}C_n}{2} \Rightarrow \sum_{r=0}^n {}^{2n}C_r = \frac{4^n + {}^{2n}C_n}{2}$$

1.(2) z corresponding to point P has minimum arg.

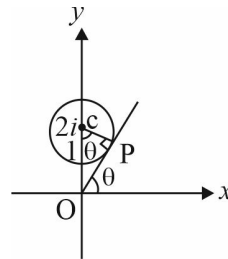
$$\therefore OP = \sqrt{(OC)^2 - (CP)^2} = \sqrt{3}$$

$$\text{In } \triangle OCP, \sin \theta = \frac{\sqrt{3}}{2}, \cos \theta = \frac{1}{2}$$

$$\therefore P \equiv OP \cos \theta + i OP \sin \theta$$

$$\Rightarrow P \equiv \frac{\sqrt{3}}{2} + i \frac{3}{2}$$

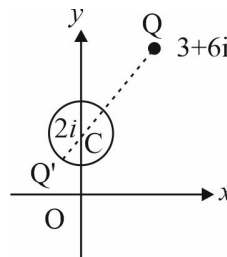
$$\frac{b}{a^2} = \left(\frac{3}{2}\right) / \left(\frac{3}{4}\right) = 2$$



2.(6) $|z - 3 - 6i|$ maximum along and QCQ' .

$$\therefore |z - 3 - 6i|_{\max} = QC + CQ'$$

$$= |3 + 4i| + 1 = 6.$$



$$3.(2) \quad A_k = \cos\left(\frac{2k\pi}{10}\right) + i \sin\left(\frac{2k\pi}{10}\right) = e^{\frac{i2k\pi}{10}}.$$

$\Rightarrow A_k$ is 10^{th} root of unity.

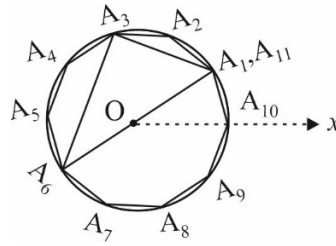
By observation

$$\text{Here } ar(\Delta A_1 A_3 A_6) = ar(\Delta A_2 A_4 A_7)$$

$$= ar(\Delta A_3 A_5 A_8) = ar(\Delta A_4 A_6 A_9)$$

$$= ar(\Delta A_5 A_7 A_{10}) = ar(\Delta A_6 A_8 A_{11})$$

$$\frac{\sum_{k=1}^6 ar(\Delta A_k A_{k+2} A_{k+5})}{\sum_{k=1}^3 ar(\Delta A_{k+1} A_{k+3} A_{k+6})} = 2$$



$$4.(2) \quad \prod_{n=1}^{10} (2 - \alpha^n) = (2 - \alpha)(2 - \alpha^2)(2 - \alpha^3) \dots (2 - \alpha^{10})$$

since α is 11^{th} root of unity

$$\therefore x^{11} - 1 = (x - 1)(x - \alpha)(x - \alpha^2) \dots (x - \alpha^{10})$$

$$\frac{x^{11} - 1}{x - 1} = (x - \alpha)(x - \alpha^2)(x - \alpha^3) \dots (x - \alpha^{10})$$

$$2^{11} - 1 = (2 - \alpha)(2 - \alpha^2)(2 - \alpha^3) \dots (2 - \alpha^{10})$$

$$\text{and } \frac{(-2)^{11} - 1}{-2 - 1} = (-2 - \alpha)(-2 - \alpha^2)(-2 - \alpha^3) \dots (-2 - \alpha^{10}) = (2 + \alpha)(2 + \alpha^2)(2 + \alpha^3) \dots (2 + \alpha^{10})$$

$$\therefore \frac{\prod_{n=1}^{10} (2 - \alpha^n)}{\prod_{n=1}^{10} (2 + \alpha^n)} = \left(\frac{2^{11} - 1}{2^{11} + 1} \right) \times 3 = \frac{2047}{2049} \times 3 \therefore \left[\frac{\prod_{n=1}^{10} (2 - \alpha^n)}{\prod_{n=1}^{10} (2 + \alpha^n)} \right] = 2$$

$$5.(15) \quad \text{No. of ways} = \text{coeff. of } x^3 \text{ in } (x^0 + x^1)(x^0 + x^1 + x^2)(x^0 + x^1 + x^2 + x^3)(x^0 + x^1 + x^2 + x^3 + x^4)$$

$$= \text{coefficient of } x^3 \text{ in } (1 + x)(1 + x + x^2)(1 + x + x^2 + x^3)(1 + x + x^2 + x^3 + x^4)$$

$$= \text{coefficient of } x^3 \text{ in } (1 + x)(1 + x + x^2)(1 + x + x^2 + x^3)(1 + x + x^2 + x^3)$$

$$= \text{coefficient of } x^3 \text{ in } (1 + x)(1 + x + x^2)(1 + x + x^2 + x^3)^2$$

$$= \text{coefficient of } x^3 \text{ in } (1 + 2x + 2x^2 + x^3)(1 + x + x^2 + x^3)^2$$

$$= \text{coefficient of } x^3 \text{ in } (1 + 2x + 2x^2 + x^3)(1 + 3x^2 + 2x + 4x^3) = 15$$

$$6.(27) \quad x_1 + x_2 + x_3 = 5 \rightarrow \text{No. of sol.} = {}^7C_5 = 21$$

$$x_1 + x_2 + x_3 = 2 \rightarrow \text{No. of sol.} = {}^4C_2 = 6$$

$$\therefore \text{Total no. of solutions} = 27$$

Solutions to JEE Advanced – 2 | JEE 2023 | Paper – 2 (Gen 1 & 2)

PHYSICS

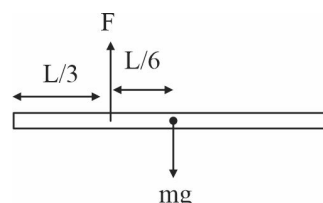
1.(BCD) Moment of inertia of the rod about the pivot,

$$I = \frac{1}{12}mL^2 + m\left(\frac{L}{2} - \frac{L}{3}\right)^2 = \frac{1}{9}mL^2$$

So, the initial angular acceleration of the rod is given by

$$\tau = I\alpha$$

$$\Rightarrow mg\left(\frac{L}{6}\right) = \left(\frac{1}{9}mL^2\right)\alpha \Rightarrow \alpha = \frac{3g}{2L}$$



$$\text{So, the initial acceleration of the CM} = \alpha\left(\frac{L}{6}\right) = \frac{g}{4}$$

Let the force applied by the pivot on the rod just after it is released be F upwards

$$\text{Then, } mg - F = m\left(\frac{g}{4}\right) \Rightarrow F = \frac{3mg}{4}$$

The rod attains its maximum angular velocity ω when it is vertical

By conservation of energy between the horizontal and vertical positions,

$$mg\left(\frac{L}{6}\right) = \frac{1}{2}\left(\frac{1}{9}mL^2\right)\omega^2 \Rightarrow \omega = \sqrt{\frac{3g}{L}}$$

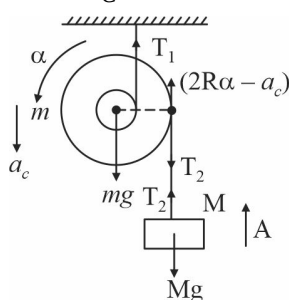
And, in the vertical position, the angular acceleration of the rod is zero. Hence, the acceleration of its CM

$$\text{is } \omega^2\left(\frac{L}{6}\right) = \frac{g}{2}$$

2.(AD) Let us assume that the acceleration of the centre of the pulley and the block are a_C and A , directed downwards and upwards respectively, and the angular acceleration of the pulley be α anti-clockwise

Then, since the strings are wound on the pulley and hence do not slip on it, we get the following equations from the two points on the pulley where the strings begin contact with it:

$$\begin{aligned} a_C &= R\alpha & \text{and} & & 2R\alpha - a_C &= A \\ \Rightarrow A &= R\alpha & \text{and} & & A &= a_C \end{aligned}$$



Since a_C and A are assumed in opposite directions, we get $\vec{a}_C = -\vec{A}$

(This relation is independent of the masses of the bodies)

Now, let the tension in the upper and lower strings be T_1 and T_2 respectively

$$\text{Then, } T_2 - Mg = MA$$

$$T_2 + mg - T_1 = ma_C$$

$$\text{And } T_1 R - T_2 (2R) = \left(\frac{3}{2} m R^2 \right) \alpha$$

Simplifying the equations, we get

$$T_2 - Mg = MR\alpha \quad \dots (i)$$

$$T_2 + mg - T_1 = mR\alpha \quad \dots (ii)$$

$$T_1 - 2T_2 = \frac{3}{2} mR\alpha \quad \dots (iii)$$

Adding all three equations, we get

$$\alpha = \frac{(m - M)g}{(5m/2 + M)R}$$

$$\text{Therefore, } a_C = A = \frac{(m - M)g}{(5m/2 + M)}$$

So, if $M = 2m$, $A = -\frac{2g}{9}$, i.e. $2g/9$ downwards

And, if $M = m/2$, $A = \frac{g}{6}$, i.e. $g/6$ upwards

3.(BD) Components of velocity upon reaching the ground for the first time:

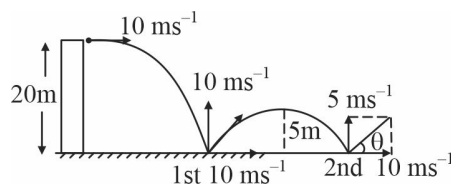
$$V_{X_0} = 10 \quad V_{Y_0} = \sqrt{2(10)(20)} = 20$$

After the first rebound, the X-component remains unchanged, and because now the velocity vector makes 45° angle with the horizontal,

$$v_{X1} = v_{Y1} = 10$$

Therefore, the maximum height reached before the ball strikes the ground again,

$$H = \frac{v_{Y1}^2}{2g} = 5 \text{ m}$$



Also, the coefficient of restitution,

$$e = \frac{v_{Y1}}{v_{Y0}} = \frac{10}{20} = \frac{1}{2}$$

Therefore, the velocity components after the second rebound are:

$$v_{X2} = 10 \quad v_{Y2} = ev_{Y1} = \frac{1}{2}(10) = 5$$

So, after the second rebound, the angle made by the velocity vector with the horizontal,

$$\theta = \tan^{-1} \left(\frac{v_{Y2}}{v_{X2}} \right) = \tan^{-1} \left(\frac{1}{2} \right)$$

4.(ABD) For $0 \leq t < 2$,

$$v(t) = v(0) + \int_0^t (3t - t^2) dt = \frac{3t^2}{2} - \frac{t^3}{3}$$

$$x(t) = x(0) + \int_0^t v(t) dt = \frac{t^3}{2} - \frac{t^4}{12}$$

For $2 \leq t < 4$,

$$v(t) = v(2) + \int_2^t (-4 + 5t - t^2) dt = \frac{10}{3} + \left(-4(t-2) + \frac{5}{2}(t^2-4) - \frac{1}{3}(t^3-8) \right)$$

$$\Rightarrow v(t) = 4 - 4t + \frac{5}{2}t^2 - \frac{1}{3}t^3$$

$$x(t) = x(2) + \int_2^t v(t) dt = -\frac{t^4}{12} + \frac{5}{6}t^3 - 2t^2 + 4t - \frac{8}{3}$$

$$\text{So, } v(1) = \frac{7}{6} \quad \text{and} \quad v(3) = \frac{11}{2}$$

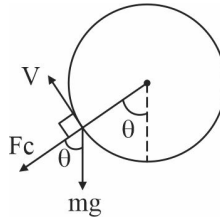
Displacement between $t = 1$ and $t = 3$ is $x(3) - x(1) = 6.66$ And, $x(4) = 13.33$

5.(AC) We can understand this quite simply using the concept of pseudo force.

If we look at the point mass in the reference frame of the rotating disc, the point mass is in equilibrium. Hence, net force on it must be zero.

We already know two forces acting on the point mass: its weight $W = mg$ acting downwards, and the pseudo force $F_C = m\omega^2 r$ (the centrifugal force) acting radially outward.

Then, the force $\vec{F}$ exerted by the disc on the mass m , must be such that it balances the resultant of the above two forces (because the point mass is in equilibrium in the frame of the disc)



$$\text{We know that } W = mg \text{ and } F_C = m\omega^2 R = m \left(\sqrt{\frac{g}{2R}} \right)^2 R = \frac{mg}{2}$$

And we know that $\vec{F} = -(\vec{W} + \vec{F}_C)$, i.e. the force exerted by the disc on the point mass m is equal and opposite to the resultant of the other two forces.

Since $\vec{W}$ always acts downward, and $\vec{F}_C$ acts radially outward, the instantaneous angle between them is θ itself

$$\text{So, } |\vec{F}| = \sqrt{W^2 + F_C^2 + 2(W)(F_C)\cos\theta} = \sqrt{(mg)^2 + \left(\frac{mg}{2}\right)^2 + 2(mg)\left(\frac{mg}{2}\right)\cos\theta}$$

$$\Rightarrow |\vec{F}| = \left(\frac{5}{4} + \cos\theta \right)^{1/2} mg$$

$$\text{So, } F\left(\frac{2\pi}{3}\right) = \left(\frac{5}{4} + \left(-\frac{1}{2}\right)\right)^{1/2} mg = \frac{\sqrt{3}}{2} mg$$

$$F(\pi) = \left(\frac{5}{4} + (-1)\right)^{1/2} mg = \frac{mg}{2}$$

We know that if $\vec{v}$ is the instantaneous velocity of the point mass, then $P(\theta) = \vec{F} \cdot \vec{v} = -(\vec{W} + \vec{F}_C) \cdot \vec{v}$

Now, since $\vec{F}_C$ is always radial and $\vec{v}$ is always tangential to the disc, $\vec{F}_C \cdot \vec{v} = 0$ at all instants

And, since the instantaneous angle between $\vec{W}$ and $\vec{v}$ is $\left(\frac{\pi}{2} + \theta\right)$,

$$P(\theta) = -\vec{W} \cdot \vec{v} = -mg(R\omega) \cos\left(\frac{\pi}{2} + \theta\right) = mgR\omega \sin \theta$$

$$\text{So, } P\left(\frac{\pi}{4}\right) = \frac{mgR\omega}{\sqrt{2}} \quad \text{and} \quad P\left(\frac{3\pi}{4}\right) = \frac{mgR\omega}{\sqrt{2}}$$

6.(ABD) Before the string breaks, the acceleration of both blocks,

$$a = \left(\frac{2.1 - 1.9}{2.1 + 1.9}\right)g = 0.5 \text{ m/s}^2 \text{ upward for block A and downward for block B}$$

Therefore, at $t = 1.0 \text{ s}$, the velocities of the blocks are: (taking upward positive)

$$v_{A1} = (0.5)(1) = 0.5 \text{ m/s}$$

$$v_{B1} = (-0.5)(1) = -0.5 \text{ m/s}$$

After the string breaks, acceleration of both blocks is g downwards

So, at $t = 1.1 \text{ s}$, the velocities of the blocks are:

$$v_A = 0.5 + (-10)(0.1) = -0.5 \text{ m/s}$$

$$v_B = -0.5 + (-10)(0.1) = -1.5 \text{ m/s}$$

Now, between $t = 0$ and $t = 1.1 \text{ s}$, the total displacements of the two blocks are:

$$s_A = \left(\frac{1}{2}(0.5)(1)^2\right) + \left((0.5)(0.1) + \frac{1}{2}(-10)(0.1)^2\right) = 0.25 \text{ m} = 25 \text{ cm}$$

$$s_B = \left(\frac{1}{2}(-0.5)(1)^2\right) + \left((-0.5)(0.1) + \frac{1}{2}(-10)(0.1)^2\right) = -0.35 \text{ m} = -35 \text{ cm}$$

7.(B) Conserving angular momentum about the hinged point,

Initial angular momentum = Final angular momentum

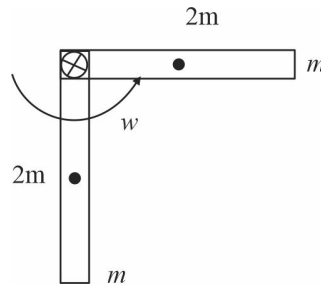
$$\Rightarrow mvL = \left(\frac{1}{3}(2m)L^2 + mL^2\right)\omega \Rightarrow \omega = \frac{3v}{5L}$$

8.(B) Moment of inertia of the combined system after the collision, about the pivot is

$$I = \left(\frac{1}{3}(2m)L^2 + mL^2\right) = \frac{5}{3}mL^2$$

Then, conserving energy,

Loss in KE = Gain in GPE



$$\frac{1}{2} \left(\frac{5}{3} mL^2 \right) \omega^2 = (2m)g \left(\frac{L}{2} \right) + mgL$$

$$\left(\because \omega = \frac{3V}{5L} \right)$$

$$V^2 = \frac{20gL}{3}$$

9.(C) The length of the spring at the instant the block comes to rest is $\sqrt{L^2 + (3L/4)^2} = 5L/4$

So, the elongation at this instant is $x = 5L/4 - L = L/4$

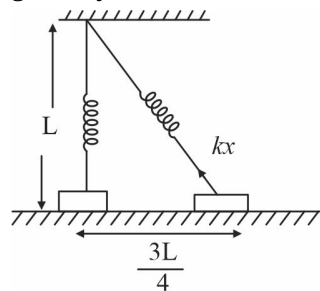
Conserving energy,

Loss in KE = Gain in elastic PE

$$\frac{1}{2} Mv^2 = \frac{1}{2} k \left(\frac{L}{4} \right)^2 \Rightarrow v = \frac{L}{4} \sqrt{\frac{k}{M}}$$

10.(A) Let the angle made by the spring with the horizontal at this instant be θ

Then, the acceleration a of the block is given by $kx \cos \theta = Ma$



$$\text{Now, } \cos \theta = \frac{3L/4}{5L/4} = 3/5$$

$$\text{So, } a = \frac{kx \cos \theta}{M} = \frac{3}{20} \frac{kL}{M}$$

11.(C) Since there is no external force on the system in the horizontal direction, and the CM of the system (which is initially at the mid-point of the rod) is initially at rest, it remains at rest

At $t = t_1$, as the rod is parallel to the Y-axis, and the CM of the system will be collinear with the rod itself, the

rod at this instant is along the line $x = \frac{L}{2}$

So, at this instant, let the coordinates of the man A (and the mid-point of the rod) be $\left(\frac{L}{2}, y \right)$

Then, the coordinates of the man B must be $\left(\frac{L}{2}, y - \frac{L}{2} \right)$

Since the CM still must be at $\left(\frac{L}{2}, 0\right)$,

$$Y_{CM} = 0 \Rightarrow \frac{M(y) + M(y) + M\left(y - \frac{L}{2}\right)}{M + M + M} = 0 \Rightarrow y = \frac{L}{6}$$

So, at $t = t_1$, the mid-point of the rod has coordinates $\left(\frac{L}{2}, \frac{L}{6}\right)$

Hence, at this instant, the man B has the coordinates $\left(\frac{L}{2}, -\frac{L}{3}\right)$

12.(B) Now, let the angular velocity of the rod at $t = t_1$ be ω

Between $t = 0$ and $t = t_1$, we can conserve the angular momentum of the system about any point in space as there is no external force, and hence no external torque about any point in space

Let us conserve the angular momentum about the CM of the system, since that is quite convenient to work out
The moment of inertia of the system about its CM:

$$\text{At } t = 0 \quad I_0 = \frac{1}{12}ML^2 + 2\left(M\left(\frac{L}{2}\right)^2\right) = \frac{7}{12}ML^2$$

$$\text{At } t = t_1 \quad I_1 = \left(\frac{1}{12}ML^2 + M\left(\frac{L}{6}\right)^2\right) + M\left(\frac{L}{6}\right)^2 + M\left(\frac{L}{6} - \frac{L}{2}\right)^2 \Rightarrow I_1 = \frac{1}{4}ML^2$$

So, about the CM of the system,

Initial angular momentum = Final angular momentum

$$I_0\omega_0 = I_1\omega_1$$

(as at both instants, the men are at relative to the rod)

$$\Rightarrow \left(\frac{7}{12}ML^2\right)\omega_0 = \left(\frac{1}{4}ML^2\right)\omega_1 \Rightarrow \omega_1 = \frac{7}{3}\omega_0$$

So, at $t = t_1$, the velocity of man A is $v_A = \omega_1\left(\frac{L}{6}\right) = \frac{7}{18}L\omega_0$

1.(2) Instantaneous position of the particle, $\vec{r} = \cos(3t)\hat{i} + 2\sin(3t)\hat{j}$

Instantaneous velocity of the particle, $\vec{v} = \frac{d}{dt}(\vec{r}) = (-3\sin(3t))\hat{i} + 6\cos(3t)\hat{j}$

Instantaneous acceleration of the particle, $\vec{a} = \frac{d}{dt}(\vec{v}) = (-9\cos(3t))\hat{i} - 18\sin(3t)\hat{j}$

Velocity and acceleration must be perpendicular

Therefore, $\vec{a} \cdot \vec{v} = 0$

$$\Rightarrow 27\sin(3t)\cos(3t) - 108\sin(3t)\cos(3t) = 0 \Rightarrow \sin(3t)\cos(3t) = 0$$

$$\Rightarrow \sin(6t) = 0 \Rightarrow t = \frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}, \dots$$

Position at $t = \frac{\pi}{6}$, $\vec{r} = \cos\left(\frac{\pi}{6}\right)\hat{i} + 2\sin\left(\frac{\pi}{6}\right)\hat{j} = 2\hat{j}$

2.(20) The time instants $t = 1$ s and $t = 4$ s must be the solutions to the equation:

$$h = ut - \frac{1}{2}(10)t^2 \quad \text{i.e.} \quad t^2 - \left(\frac{u}{5}\right)t + \frac{h}{5} = 0$$

Therefore, the product of the roots of the quadratic equation, i.e. $\frac{h}{5}$ must be equal to $(1)(4) = 4$

Therefore, $h = 20$ m

3.(4) We are given that $y = 4\sqrt{x} \Rightarrow y^2 = 16x$

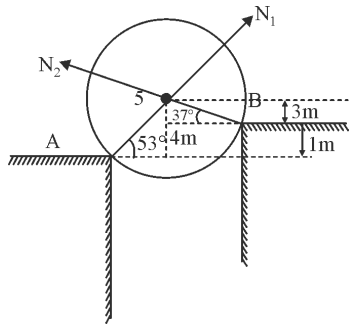
Differentiating with respect to time, $2y \frac{dy}{dt} = 16 \frac{dx}{dt} \Rightarrow yv_y = 8v_x$

If $x = 4$, $y = 8 \Rightarrow v_y = v_x$

Now, we are also given that $\sqrt{v_x^2 + v_y^2} = 4\sqrt{2}$

Therefore, at this instant, $v_y = v_x = 4$

4.(32) Let N_1 and N_2 denote the forces acting on the sphere from the supports A and B respectively



$$\begin{array}{c} \uparrow N_2 \sin 37^\circ = \frac{3N_2}{5} \\ \uparrow N_1 \sin 53^\circ = \frac{4N_1}{5} \\ \leftarrow N_2 \cos 37^\circ = \frac{4N_2}{5} \quad \rightarrow N_1 \cos 53^\circ = \frac{3N_1}{5} \\ \downarrow 100 \end{array}$$

Balancing forces,

$$\frac{3N_1}{5} = \frac{4N_2}{5} \quad \dots(i)$$

$$\frac{4N_1}{5} + \frac{3N_2}{5} = 40 \quad \dots(ii)$$

Solving, we get

$$N_1 = 32 \text{ N}$$

$$N_2 = 24 \text{ N}$$

5.(1.25)

Since the force is always perpendicular to the velocity of the particle, it cannot change the speed of the particle, and will only change its direction of motion

The instantaneous radial acceleration of the particle is $a_r = \frac{F}{m} = \frac{2t}{1} = 2t$

Now we know that if R is the instantaneous radius of curvature of the path of the particle, the radial acceleration is $\frac{v^2}{R} = \omega^2 R$

Since we need to find the angle turned by the particle, we need to find its instantaneous angular velocity (and we do not need the radius of curvature), we write the radial acceleration as

$$a_r = \omega^2 R = \omega^2 \left(\frac{v}{\omega} \right) = v\omega$$

$$\Rightarrow \quad \omega = \frac{a_r}{v} = \frac{2t}{20} = \frac{t}{10} \quad \Rightarrow \quad \theta = \int_0^5 \omega dt = \frac{t^2}{20} \Big|_0^5 = 1.25 \text{ rad}$$

6.(0.80) Let the velocity of the CM at the end of the rough portion be v_1

Conserving energy in the top half,

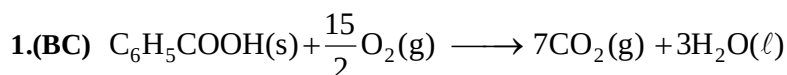
$$mg \frac{h}{2} = \frac{1}{2} m v_1^2 + \frac{1}{2} (mk^2) \left(\frac{v_1^2}{R^2} \right) \quad \dots (i)$$

Conserving energy in the bottom half,

$$mg \frac{h}{2} = \frac{1}{2} m \left(\left(\sqrt{\frac{14gh}{9}} \right)^2 - v_1^2 \right) \quad \Rightarrow \quad v_1^2 = \frac{5gh}{9}$$

Using this in equation (i), we get $\frac{k^2}{R^2} = \frac{4}{5} = 0.8$

CHEMISTRY



$$\Delta_{\text{C}}H = 7(-390) + 3(-285) - (-410) = -3175 \text{ kJ/mol}$$

$$\Delta_{\text{C}}U = \Delta_{\text{C}}H - \Delta n_{\text{g}}RT = -3175 - \left(\frac{-1}{2}\right) \times \frac{8.314 \times 300}{1000} = -3173.75 \text{ kJ/mol}$$

2.(BD) Mass of solution = 900 + 138 = 1038 g

Volume of solution = 1 L

$$\text{Density of solution} = \frac{1038}{1000} = 1.038 \frac{\text{g}}{\text{cm}^3} = 1038 \frac{\text{kg}}{\text{m}^3}$$

$$X_{\text{C}_2\text{H}_5\text{OH}} = \frac{3}{3+50} = \frac{3}{53}$$

$$M = \frac{3}{1} = 3 \text{ mol/L}$$

$$m = \frac{3 \times 1000}{900} = \frac{10}{3} \text{ mol/kg}$$

3.(BCD) (B) Real gas behaves ideally at boyle's temperature

(C) $T_{\text{B}} = \frac{a}{Rb}$

(D) $Z = 1 - \frac{2a}{VRT}$ for 2 moles of vander waal's gas at low pressure

4.(BC) $4.25 = \phi_{\text{A}} + T_{\text{A}}$

$$4.2 = \phi_{\text{B}} + (T_{\text{A}} - 2.5)$$

$$0.05 = \phi_{\text{A}} - \phi_{\text{B}} + 2.5$$

$$\phi_{\text{B}} - \phi_{\text{A}} = 2.45$$

$$\lambda_{\text{B}} = 2\lambda_{\text{A}}$$

$$\frac{h}{\sqrt{2mT_{\text{B}}}} = \frac{2h}{\sqrt{2mT_{\text{A}}}} \Rightarrow T_{\text{A}} = 4T_{\text{B}}$$

$$T_{\text{A}} = 4(T_{\text{A}} - 2.5)$$

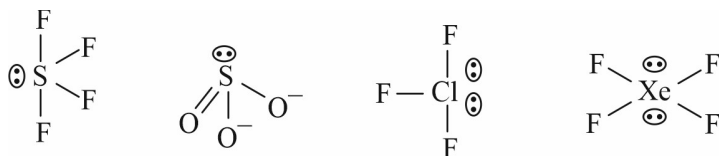
$$3T_{\text{A}} = 10 \Rightarrow T_{\text{A}} = \frac{10}{3} \text{ eV}$$

$$T_{\text{B}} = \frac{5}{6} \text{ eV}$$

$$4.25 = \phi_{\text{A}} + \frac{10}{3} \Rightarrow \phi_{\text{A}} = 0.92 \text{ eV}$$

$$\phi_{\text{B}} = 3.37 \text{ eV}$$

5.(AD)



6.(BCD) (B) Ionic radius $\Rightarrow O^{2-} > F^- > Na^+ > Mg^{2+}$

(C) $Li < B < Be < C (IE_1)$

(D) $Cl > F > Br > I$ (EA)

7.(B) Stability of $N_2^+ > N_2^-$ so bond energy of $N_2^+ > N_2^-$

8.(B) Bond order of $N_2^+, O_2^+ = 2.5$

Bond order of $F_2 = 1$

Bond order $C_2^+ = 1.5$

No. of ABMO $e^- \Rightarrow O_2^+ > N_2^+$

9.(D) I.E $\Rightarrow He > Ne > Ar > Kr$

10.(B) $Na^- \xrightarrow{\Delta H = x} Na \xrightarrow{\Delta H = y} Na^+$
 $y > x$

11.(B) $w = -nR \cdot \Delta T = -1 \times 8.314 \times 72 = -598.6 J$
 $= -0.6 kJ$

12.(A) $\Delta U = q + w = 1.6 + (-0.6) = 1.0 kJ$

1.(4) $PCl_3F_2, CCl_4, SF_6, C_2H_4$ have zero dipole moment.

2.(2) $IE_1 < IE_2 \ll IE_3 < IE_4 \Rightarrow +2$ state is more stable $\Rightarrow 2^{nd}$ group

3.(15) $\Delta G^\circ = -RT \ln K_p$

$\Delta H^\circ - T\Delta S^\circ = -RT \ln K_p$

$(10.5 \times 10^3) - 300 \Delta S^\circ = -2 \times 300(-10)$

$\Delta S^\circ = \frac{4500}{300} = 15$

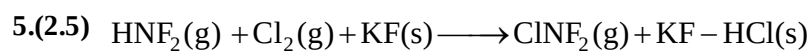
4.(5) Radial Probability density $= R^2(r)$

at $r = 0 \Rightarrow R^2(r) = \frac{1}{\pi} \left(\frac{1}{a_0} \right)^3$

at $r = 2a_0 \Rightarrow R^2(r) = \frac{1}{\pi} \left(\frac{1}{a_0} \right)^3 e^{-4}$

$\frac{R^2(2a_0)}{R^2(0)} = e^{-4} = x e^{-y}$

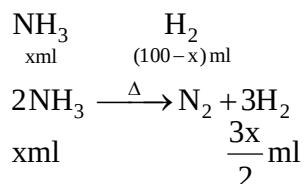
$x = 1, y = 4 \Rightarrow x + y = 5$



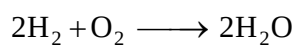
$$\begin{array}{ccc} a \text{ mmol} & a \text{ mmol} & \\ \frac{2a}{3} \text{ mmol} & \frac{2a}{3} \text{ mmol} & \frac{a}{3} \text{ mmol} \end{array}$$

$$\text{Volume percent of ClNF}_2 \text{ in gaseous mixture} = \frac{a/3}{5a/3} \times 100 = 20\%$$

6.(0.96)



$$\text{Volume of H}_2 = \frac{3x}{2} + 100 - x = 100 + \frac{x}{2}$$



$$\text{Volume of O}_2 \text{ used} = \left(50 + \frac{x}{4}\right) \text{ ml}$$

$$80 - \left(50 + \frac{x}{4}\right) = 6$$

$$24 = \frac{x}{4} \Rightarrow x = 96$$

fraction of NH_3 in original mixture = 0.96

MATHEMATICS

1.(ABD) Given expression $= (3 - 5\omega + 7\omega^2)^P (1 + \omega^{2P} + \omega^P) = 0$

$$\Rightarrow p = 3n+1, 3n+2, n \in I$$

2.(ABC) Let $\arg z = \theta$ then $z = e^{i\theta} \quad \therefore \frac{1+z^2}{1+\bar{z}^2} = \frac{1+e^{i2\theta}}{1+e^{-i2\theta}} = e^{i2\theta}$

$$\arg\left(\frac{1+z^2}{1+\bar{z}^2}\right) = 2\theta \in \left[\frac{\pi}{3}, \frac{2\pi}{3}\right]$$

3.(AC) By observation, $x=1$ gives same relation in both equation.

$$\therefore \text{common root} = x=1 \text{ when } 1+a+b+c=0 \text{ or } a+b+c=-1$$

4.(ABD) $(1+2x+3x^2)^{50} = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_{100}x^{100} \quad \dots(1)$

Put $x=0$ in (1), we get $a_0 = 1$

Also, put $x=1$ in (1) to get $6^{50} = a_0 + a_1 + a_2 + \dots + a_{100}$

Differentiate (1) w.r.t. x , we get

$$50(1+2x+3x^2)^{49}(2+6x) = a_1 + a_2(2x) + a_3(3x^2) + \dots + a_{100}(100x^{99}) \quad \dots(2)$$

Put $x=0$ in (2) to get, $a_1 = 100$

Differentiate (2) w.r.t. x , to get

$$50 \times 49(1+2x+3x^2)^{48}(2+6x)^2 + 300(1+2x+3x^2)^{49} \\ = 2 \cdot a_2 + a_3(6x) + a_4(12x^2) + \dots + a_{100}(9900x^{98})$$

Put $x=0$ to get, $a_2 = 5050$

5.(ABCD) (A) Fundamental period = $\frac{\pi}{2}$

(B) Fundamental period = π

(C) Fundamental period = $\frac{\pi}{2}$

(D) Not periodic

6.(ABD) No. of ways to invite one or more friends

$$= {}^3C_1 + {}^3C_2 + {}^3C_3 = 7$$

No. of zeroes at the end of $\underline{31}$ = exponent of 5 in $\underline{31} = \left[\frac{31}{5}\right] + \left[\frac{31}{5^2}\right] = 7$

7.(C) $(1+x)^{101} = {}^{101}C_0 + {}^{101}C_1x + {}^{101}C_2x^2 + {}^{101}C_3x^3 + {}^{101}C_4x^4 \\ + {}^{101}C_5x^5 + {}^{101}C_6x^6 + \dots + {}^{101}C_{101}x^{101} \quad \dots(1)$

Differentiate (1) w.r.t. x to get

$$101(1+x)^{100} = {}^{101}C_1 + {}^{101}C_2(2x) + {}^{101}C_3(3x^2) + {}^{101}C_4(4x^3) + {}^{101}C_5(5x^4) + \dots \\ + {}^{101}C_{101}(101x^{100}) \quad \dots(2)$$

Put $x=1, \omega, \omega^2$ respectively in (2) and add to get,

$$101(2^{100}) + 101(1+\omega)^{100} + 101(1+\omega^2)^{100}$$

$$\begin{aligned}
 &= {}^{101}C_1(3) + {}^{101}C_2 2(1 + \omega + \omega^2) + {}^{101}C_3(3 + 3\omega^2 + 3\omega^4) + {}^{101}C_4(4 + 4\omega^3 + 4\omega^6) + {}^{101}C_5(5 + 5\omega^4 + 5\omega^8) + \dots \\
 &+ {}^{101}C_{100}(100 + 100\omega^{99} + 100\omega^{198}) + {}^{101}C_{101}(101 + 101\omega^{100} + 101\omega^{200}) \\
 \Rightarrow &101(2^{100}) + 101\omega^{200} + 101\omega^{100} \\
 &= {}^{101}C_1(3) + {}^{101}C_2 2(0) + {}^{101}C_3 3(0) + {}^{101}C_4(4) + {}^{101}C_5(0) + \dots + {}^{101}C_{100}100(3.100) \\
 \Rightarrow &101.2^{100} + 101(\omega^2 + \omega) = 3 \left[{}^{101}C_1 + 4 {}^{101}C_4 + 7. {}^{101}C_7 + \dots + 100. {}^{101}C_{100} \right] \\
 \Rightarrow &\frac{101(2^{100} - 1)}{3} = {}^{101}C_1 + 4. {}^{101}C_4 + 7. {}^{101}C_7 + \dots + 100. {}^{101}C_{100}
 \end{aligned}$$

8.(A) Put $x = i$ and $-i$ respectively in (2) and add to get,

$$\begin{aligned}
 &101(1+i)^{100} + 101(1-i)^{100} = {}^{101}C_1(2) + {}^{101}C_2(2i - 2i) + {}^{101}C_3(3i^2 + 3i^2) + {}^{101}C_4(4i^3 - 4i^3) \\
 &+ {}^{101}C_5[5i^4 + 5i^4] + \dots + {}^{101}C_{101}(101i^{100} + 101i^{100}) \\
 \Rightarrow &101 \left[(2i)^{50} + (2i)^{50} \right] = 2 \left[{}^{101}C_1 - 3. {}^{101}C_3 + 5. {}^{101}C_5 - 7. {}^{101}C_7 + \dots + 101. {}^{101}C_{101} \right] \\
 \Rightarrow &-2^{50}.101 = {}^{101}C_1 - 3. {}^{101}C_3 + 5. {}^{101}C_5 - 7. {}^{101}C_7 + \dots + 101. {}^{101}C_{101}.
 \end{aligned}$$

9.(D) $f(x) + 2f(1-x) = x^2 + 2 \quad \forall x \dots (1)$

Replace x by $1-x$ in (1) to get $f(1-x) + 2f(x) = (1-x)^2 + 2 \dots (2)$

Operate $2 \times (2) - (1)$ to get $f(x) = \frac{1}{3} \left[2 + 2(1-x)^2 - x^2 \right]$

$$\Rightarrow f(x) = \frac{1}{3}(x-2)^2 \dots (3)$$

Similarly, $2g(x-1) - g\left(\frac{1}{x}-1\right) = x \dots (4)$

Replace x by $\frac{1}{x}$ in (4) to get $2g\left(\frac{1}{x}-1\right) - g(x-1) = \frac{1}{x} \dots (5)$

Operate $2 \times (4) + (5)$ to get $g(x-1) = \frac{1}{3} \left(2x + \frac{1}{x} \right)$

$$\therefore g(x) = \frac{1}{3} \left[\frac{2(x+1)^2 + 1}{x+1} \right] = \frac{1}{3} \left(\frac{2x^2 + 4x + 3}{x+1} \right) \dots (6)$$

From (3), $f(-1) = 3, f(5) = 3$

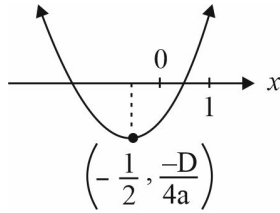
From (6), $g(0) = 1, g(1) = \frac{3}{2}$

$$\therefore 2f(-1) + 3g(0) + 4g(1) = 15 = 5f(5)$$

10.(B) $f(x) = \frac{1}{3}(x-2)^2 \geq 0$

$\therefore f(x)$ is minimum at $x = 2$.

11.(D) $\frac{-b}{2a} = \frac{-1}{2} \Rightarrow a = b$

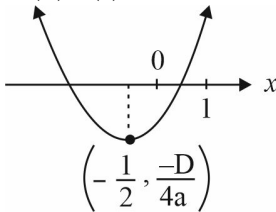


$$f(0)x^2 + f(1)x + f(-1) = 0$$

$$\Rightarrow x^2 + (a+b+1)x + (a-b+1) = 0$$

$$\Rightarrow x^2 + (2a+1)x + 1 = 0 \Rightarrow g(x) = 0$$

$$g(0)g(1) < 0$$



$$\Rightarrow 2a+3 < 0 \quad \boxed{a < -\frac{3}{2}} \quad \text{But } a > 0$$

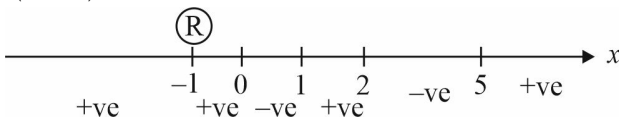
12.(B) $ax^2 + ax + 1 = 0$

$$\text{For real and distinct roots, } D > 0 \Rightarrow a^2 - 4a > 0 \Rightarrow a \in (-\infty, 0) \cup (4, \infty)$$

$$\text{But } a > 0$$

$$\therefore a \in (4, \infty)$$

1.(8) $\frac{(|x|-1)(|x-1|-1)}{(e^x+1)(|x-2|-3)} \leq 0$



$$\Rightarrow \frac{(|x|-1)(|x-1|-1)}{|x-2|-3} \leq 0 \text{ as } e^x + 1 > 0 \quad \therefore x \in [0, 1] \cup [2, 5)$$

$$a+b+c+d=8$$

2.(2) $x^4 - x^3 - 2x - 4 = (x-2)(x^3 + x^2 + 2x + 2) = (x-2)(x+1)(x^2 + 2) = 0$

$$\Rightarrow x = 2, -1, \pm \sqrt{2}i$$

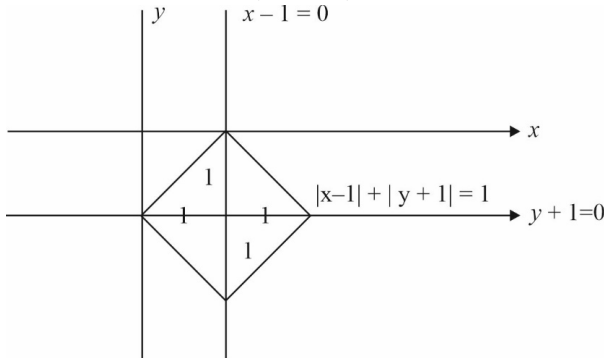
3.(1) $111^{11} + 111^{12} + 111^{13} + 111^{14} + 111^{15} + 111^{16}$
 $= (110+1)^{11} + (110+1)^{12} + (110+1)^{13} + (110+1)^{14} + (110+1)^{15} + (110+1)^{16}$
 $= \left({}^{11}C_0(110)^{11} + {}^{11}C_1(110)^{10} + {}^{11}C_2(110)^9 + \dots + {}^{11}C_{10}110 + {}^{11}C_{11} \right)$
 $+ \left({}^{12}C_0(110)^{12} + {}^{12}C_1(110)^{11} + \dots + {}^{12}C_{11}(110) + {}^{12}C_{12} \right)$
 $+ \left({}^{13}C_0(110)^{13} + {}^{13}C_1(110)^{12} + \dots + {}^{13}C_{12}(110) + {}^{13}C_{13} \right)$
 $+ \left({}^{14}C_0(110)^{14} + {}^{14}C_1(110)^{13} + \dots + {}^{14}C_{13}(110) + {}^{14}C_{14} \right)$

$$\begin{aligned}
 &+ \left({}^{15}C_0(110)^{15} + {}^{15}C_1(110)^{14} + \dots + {}^{15}C_{14}(110) + {}^{15}C_{15} \right) \\
 &+ \left({}^{16}C_0(110)^{16} + {}^{16}C_1(110)^{15} + \dots + {}^{16}C_{15}(110) + {}^{16}C_{16} \right) \\
 &\equiv 100N + \left({}^{11}C_{10} + {}^{12}C_{11} + {}^{13}C_{12} + {}^{14}C_{13} + {}^{15}C_{14} + {}^{16}C_{15} \right) 110 \\
 &+ \left({}^{11}C_{11} + {}^{12}C_{12} + {}^{13}C_{13} + {}^{14}C_{14} + {}^{15}C_{15} + {}^{16}C_{16} \right) \text{ where } N \text{ is Natural number} \\
 &\equiv 100N + (11+12+13+14+15+16)110 + 6 \\
 &\equiv 100N + 8916 \\
 \therefore \text{ digit at tenth place} &= 1.
 \end{aligned}$$

4.(9) $55 + 4\sqrt{221}i = (2\sqrt{17} + \sqrt{13}i)^2$

ALTHOUGH we can direct use the property: $|\sqrt{z}| = \sqrt{|z|}$ and thus find modulus without determining the square roots.

5.(2) Area bounded $= 4 \times \left(\frac{1}{2} \times 1 \times 1 \right) = 2$ sq. units.



6.(2164) Letters in alphabetical order $A C C E I N V$

No. of words starting with $A = \frac{|6|}{|2|} = 360$

No. of words starting with $C = \frac{|6|}{|2|} = 720$

No. of words starting with $E = \frac{|6|}{|2|} = 360$

No. of words starting with $I = \frac{|6|}{|2|} = 360$

No. of words starting with $N = \frac{|6|}{|2|} = 360$

Now words starting with V

No. of words starting with $V A C C E = |2|$

No. of words starting with $V A C C I E N = |1|$

No. of words starting with $V A C C I N E = 1$

Rank = $\frac{\quad}{2164}$